AUDIT REPORT

last-mile-dispatch: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/last-mile-dispatch
Numbers taken from commit	`c7de8a6ad78ef3961f773e55ae9af197085c3f9c` (outputs last changed in `c7de8a6 (2026-10-13 17:55:02 +0400)`)
Video	`docs/demo/demo.mp4` · 84.6 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:19
Automated checks	60 pass · 0 fail · 1 manual (of 61 figures)
Assumptions	13 (A-01 to A-13)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                  # data/ (seed 7) and outputs/*.csv, summary.json, run_log.txt (~3 min)
python3 docs/demo/source/tools/export_series.py docs/demo/source/assets/run_series.json
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_rider_board.py docs/demo/source/assets/rider_board.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/rider_board.json docs/demo/source/compositions/value.html --name SHEET
for f in s02-hook s04-routes s07-end; do python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/run_series.json docs/demo/source/compositions/$f.html --name SERIES; done
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/last-mile-dispatch-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/last-mile-dispatch-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem', 'Insights and learning'	README text	-	general knowledge	✓ verified
F-03	26 hubs	02, 05, 07	outputs/summary.json; data/hubs.csv	hubs; row count	-	direct	✓ verified
F-04	473 riders	02, 05	data/riders.csv	row count	-	direct	✓ verified
F-05	477,868 orders	02, 05	data/orders.csv.gz	row count	-	direct	✓ verified
F-06	6 weeks / 42 days	02, 05	data/days.csv	row count = 42	42 / 7 = 6 weeks	derived	✓ verified
F-07	30 minutes to each door / 30-minute promise	02, 07	src/last_mile_dispatch/config.py:27	SLA_MIN = 30	-	direct	✓ verified
F-08	a wave every 2 min / EVERY 2 MINUTES / dispatch wave 2 min	02, 03, 05	src/last_mile_dispatch/config.py:23	WAVE_MIN = 2	-	direct	✓ verified
F-09	Ramadan · weekends · rain	02	data/days.csv	ramadan, weekend, rain columns	any ramadan, any weekend, exactly one rain day	direct	✓ verified
F-10	36 riders sometimes skip doors	02	data/riders.csv	remote_marker == True	count	direct	✓ verified
F-11	hub dots + circles	02, 10	data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0)	lat/lon -> km from the centroid (demo assets/run_series.json)	-	direct	✓ verified
F-12	ORDERS → BAG → MATCH → DOOR CHECK → A/B TEST → VERDICT	03	src/last_mile_dispatch/	functions	-	direct	✓ verified
F-13	4 WEEKS	03	outputs/summary.json	experiment_days = 28	28 / 7 = 4	derived	✓ verified
F-14	cheaper orders / fewer cancels / fewer fake marks	03	outputs/ab_readout.csv	welch_hi for cost_per_delivered, cancel_rate, fake_accepted_per_1k	welch_hi < 0 for all three	derived	✓ verified
F-15	3 steps / ≤ 3 stops · 1 km	04, 05	src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67	BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0	-	direct	✓ verified
F-16	9.71	04	outputs/ablation.csv	policy=baseline, cost_per_delivered	-	direct	✓ verified
F-17	8.95 AED	04	outputs/ablation.csv	policy=bagging, cost_per_delivered	-	direct	✓ verified
F-18	generate_data.py, runner.py run_all(), simulate.py simulate_hub_day(), bagging.py, assignment.py (min_cost, fifo), geo.py, geofence.py, experiment.py, report.py	05	src/last_mile_dispatch/, scripts/	defs	-	direct	✓ verified
F-19	power.csv, ab_readout.csv, geofence_eval.csv, ablation.csv, rerandomisation.csv, hub_day_metrics.csv	05	outputs/	files	-	direct	✓ verified
F-20	3,640 rows	05	outputs/hub_day_metrics.csv	row count	26 x 28 x 5	direct	✓ verified
F-21	11,566 marks	05	outputs/geofence_eval.csv	slice=all, events	-	direct	✓ verified
F-22	hub-day runs 4,004	05	outputs/run_log.txt line 2	'simulated 4,004 hub-day-policy runs'	26 x 14 + 26 x 28 x 5 = 4,004	direct	✓ verified
F-23	full run, laptop 180 s	05	outputs/summary.json	runtime_s = 179.8	rounded to whole seconds	derived	✓ verified
F-24	cancel if not out 45 min	05	src/last_mile_dispatch/config.py:28	CANCEL_AFTER_MIN = 45	-	direct	✓ verified
F-25	150 m geofence	05	src/last_mile_dispatch/config.py:79	GEOFENCE_RADIUS_M = 150.0	-	direct	✓ verified
F-26	500 re-draws	05	outputs/rerandomisation.csv	draws	-	direct	✓ verified
F-27	13 vs 13 hubs	05, 08, 09	outputs/summary.json	arms.treatment, arms.control	len of each list	direct	✓ verified
F-28	4 parallel workers · 5 policies, 1 seed	05	src/last_mile_dispatch/cli.py:29; runner.py:36-41;	--workers default 4; same seed for every policy; ALL_POLICIES	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
			config.py:62				
F-29	CI slice 8 hubs × 7 d	05	src/last_mile_dispatch/cli.py:38, :40; .github/workflows/ci.yml	hubs.head(8), experiment days head(7)	-	direct	✓ verified
F-30	generate_data.py · seed 7	05	scripts/generate_data.py:168	--seed default 7	-	direct	✓ verified
F-31	two lines, Feb 2 - Mar 1	06	outputs/hub_day_metrics.csv	sum(cancelled) / sum(orders) by date and policy	injected SERIES block must equal the recomputation	derived	✓ verified
F-32	RAIN DAY · 42.5% → 14.0%	06	outputs/stress_split.csv	day_type=rain day, cancel_rate	x 100	direct	✓ verified
F-33	37,530 → 11,063	06	outputs/hub_day_metrics.csv	sum(cancelled) by policy	-	direct	✓ verified
F-34	−70.5%	06	outputs/hub_day_metrics.csv	F-33	(11,063 / 37,530 - 1) x 100 = -70.52%	derived	✓ verified
F-35	125,715 orders · 9.71 → 9.36 · −3.7%	06	outputs/stress_split.csv	day_type=ordinary weekday	cost_treat / cost_base - 1	derived	✓ verified
F-36	184,449 orders · 9.65 → 8.30 · −14.0%	06	outputs/stress_split.csv	day_type=weekend or Ramadan	cost_treat / cost_base - 1	derived	✓ verified
F-37	16,817 orders · 10.42 → 6.84 · −34.4%	06	outputs/stress_split.csv	day_type=rain day	cost_treat / cost_base - 1	derived	✓ verified
F-38	RAMADAN band from Feb 18	06	data/days.csv	first date with ramadan = True	-	direct	✓ verified
F-39	Rider 01-05 rows	07	demo assets/rider_board.json (re-run of the repo's simulator), injected into compositions/value.html	rows	km = sum of road legs; waited = wave time - oldest order's creation; due = earliest creation + 30 min	direct	✓ verified
F-40	795 of 821 delivered, 7 cancelled · FIFO would cancel 59	07	outputs/hub_day_metrics.csv	hub_id=18, date=2026-02-14, policy in {bagging+matching+geofence, baseline}	-	direct	✓ verified
F-41	+ 3 bags held for the next wave · 5 free riders, 8 bags waiting	07	demo assets/rider_board.json	bags_held, free_riders, bags_waiting_before	held = bags waiting - riders matched	direct	✓ verified
F-42	All 9 inside the 30-minute promise.	07	demo assets/rider_board.json	orders_dispatched; stops[].on_time	every stop ETA - creation <= 30 min	direct	✓ verified
F-43	HUB 18	07	outputs/summary.json	arms.treatment	-	direct	✓ verified
F-44	−10.6%	08	outputs/ab_readout.csv	metric=cost_per_delivered, rel_diff	x 100	direct	✓ verified
F-45	AED 9.79 → 8.75	08	outputs/ab_readout.csv	metric=cost_per_delivered, control, treatment	-	direct	✓ verified
F-46	+28.3	08	outputs/ab_readout.csv	metric=on_time_share_of_orders, diff	x 100	direct	✓ verified
F-47	58.0% → 86.2%	08	outputs/ab_readout.csv	metric=on_time_share_of_orders, control, treatment	x 100	direct	✓ verified
F-48	3.7% · control 11.6%	08	outputs/ab_readout.csv	metric=cancel_rate, treatment, control	x 100	direct	✓ verified
F-49	−72.4%	08	outputs/ab_readout.csv	metric=fake_accepted_per_1k, rel_diff	x 100	direct	✓ verified
F-50	11.7 → 3.2 per 1,000	08	outputs/ab_readout.csv	metric=fake_accepted_per_1k, control, treatment	-	direct	✓ verified
F-51	326,981 orders	08	outputs/summary.json	orders_in_experiment	-	direct	✓ verified
F-52	Welch 95% CIs exclude zero on all four	08	outputs/ab_readout.csv	welch_lo, welch_hi	lo > 0 or hi < 0 for cost, on-time, cancel, fake	derived	✓ verified
F-53	0.667 / 0.650	09	outputs/ab_readout.csv	metric=utilization, control, treatment	-	direct	✓ verified
F-54	−0.040 to +0.007	09	outputs/ab_readout.csv	metric=utilization, welch_lo, welch_hi	-	direct	✓ verified
F-55	p = 0.16	09	outputs/ab_readout.csv	metric=utilization, welch_p	-	direct	✓ verified
F-56	27 hubs per arm	09	outputs/power.csv	metric=utilization, hubs_per_arm_needed	-	direct	✓ verified
F-57	axis −0.05..+0.05 = 1,000 px; bar 95 → 571 px, estimate at 333 px	09	compositions/s06-twist.html; outputs/ab_readout.csv	welch_lo, welch_hi, diff	px = (v + 0.05) x 10,000	derived	✓ verified
F-58	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-59	~3 min	10	outputs/summary.json	runtime_s = 179.8	179.8 / 60 = 3.0 min	derived	✓ verified
F-60	SciPy · pandas · NumPy · Python	10	pyproject.toml dependencies	numpy, pandas, scipy	-	direct	✓ verified
F-61	$ make run # 26 hubs, 4 weeks, ~3 min	10	Makefile; outputs/summary.json	run target; hubs; experiment_days	-	direct	✓ verified

2. Automated check run (2026-10-14 14:19)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	26	26	✓ pass
F-04	473	473	✓ pass
F-05	477,868	477,868	✓ pass
F-06	42 days	42 days	✓ pass
F-07	true	true	✓ pass
F-08	true	true	✓ pass
F-09	true	true	✓ pass
F-10	36	36	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	4 weeks	4 weeks	✓ pass
F-14	true	true	✓ pass
F-15	true	true	✓ pass
F-16	9.71	9.71	✓ pass
F-17	8.95	8.95	✓ pass
F-18	true	true	✓ pass
F-19	true	true	✓ pass
F-20	3,640	3,640	✓ pass
F-21	11,566	11,566	✓ pass
F-22	true	true	✓ pass
F-23	180 s	180 s	✓ pass
F-24	true	true	✓ pass
F-25	true	true	✓ pass
F-26	500	500	✓ pass
F-27	13 vs 13	13 vs 13	✓ pass
F-28	true	true	✓ pass
F-29	true	true	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	42.5% → 14.0%	42.5% → 14.0%	✓ pass
F-33	37,530 → 11,063	37,530 → 11,063	✓ pass
F-34	-70.5%	-70.5%	✓ pass
F-35	125,715 · 9.71 → 9.36 · -3.7%	125,715 · 9.71 → 9.36 · -3.7%	✓ pass
F-36	184,449 · 9.65 → 8.30 · -14.0%	184,449 · 9.65 → 8.30 · -14.0%	✓ pass
F-37	16,817 · 10.42 → 6.84 · -34.4%	16,817 · 10.42 → 6.84 · -34.4%	✓ pass
F-38	2026-02-18	2026-02-18	✓ pass
F-39	true	true	✓ pass
F-40	795 of 821 delivered, 7 cancelled · FIFO 59	795 of 821 delivered, 7 cancelled · FIFO 59	✓ pass
F-41	3 held · 5 riders, 8 bags	3 held · 5 riders, 8 bags	✓ pass
F-42	true	true	✓ pass
F-43	true	true	✓ pass
F-44	-10.6%	-10.6%	✓ pass
F-45	9.79 → 8.75	9.79 → 8.75	✓ pass
F-46	+28.3	+28.3	✓ pass
F-47	58.0% → 86.2%	58.0% → 86.2%	✓ pass
F-48	3.7% · control 11.6%	3.7% · control 11.6%	✓ pass
F-49	-72.4%	-72.4%	✓ pass
F-50	11.7 → 3.2	11.7 → 3.2	✓ pass
F-51	326,981	326,981	✓ pass
F-52	true	true	✓ pass
F-53	0.667 / 0.650	0.667 / 0.650	✓ pass
F-54	-0.040 to +0.007	-0.040 to +0.007	✓ pass
F-55	0.16	0.16	✓ pass
F-56	27	27	✓ pass
F-57	true	true	✓ pass
F-58	true	true	✓ pass
F-59	~3 min	~3 min	✓ pass
F-60	true	true	✓ pass
F-61	true	true	✓ pass

3. Assumptions register

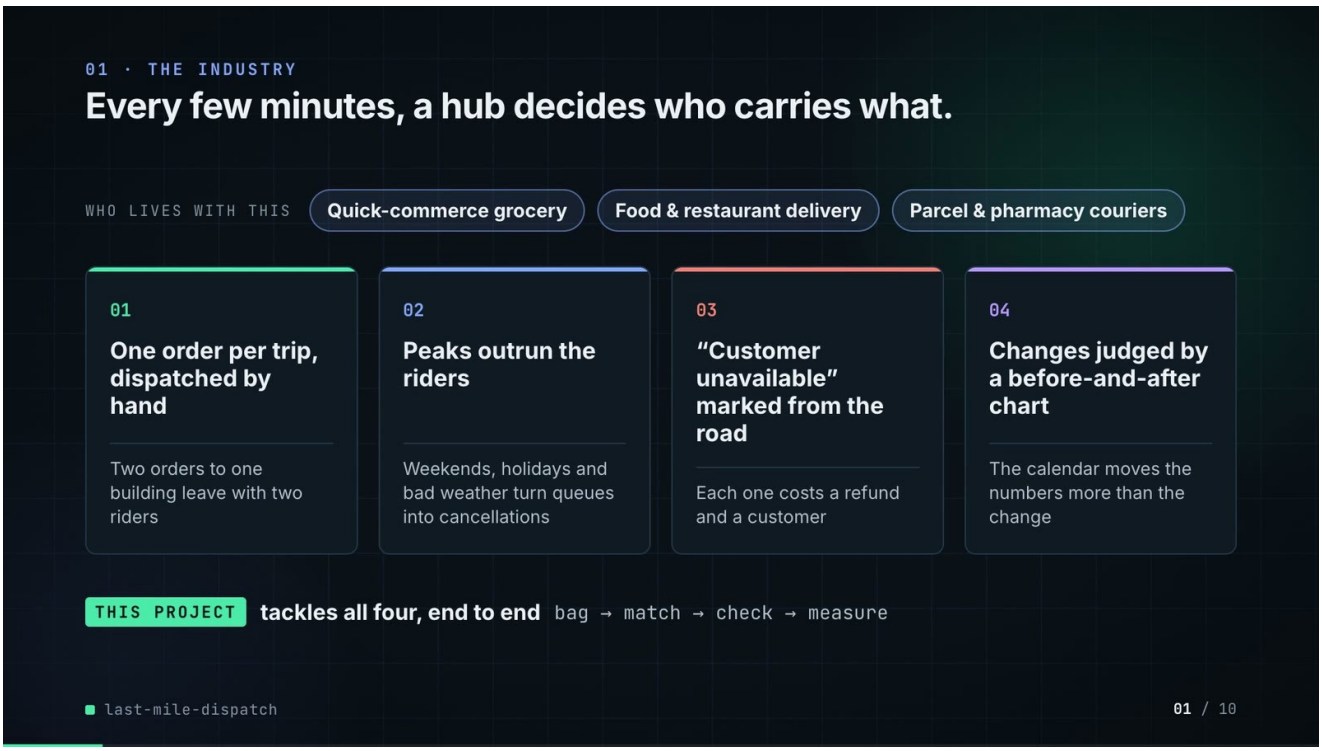
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (26 hubs, 473 riders, 42 days, 477,868 orders). Production dispatch data is proprietary.	scripts/generate_data.py:9, :168 (seed 7); src/last_mile_dispatch/config.py:11-18; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	The rider roster is fixed and sized for an ordinary weekday evening; every policy pays for the same	scripts/generate_data.py:84-85 (make_riders); README 'Known	F-04, F-33, F-34, F-35, F-36, F-37, F-44, F-45, F-53, F-54,	slides 02, 05, 06, 08, 09

ID	Assumption	Source	Figures affected	Slides affected
	riders. Cost savings therefore come mostly from cancelled orders turned into deliveries, and utilisation falls.	`limitations' (the biggest gap)`	F-55, F-56	
A-03	Travel is straight-line distance x 1.35 at an hour-dependent speed (bike 24 km/h, car 22 km/h; peaks x0.68-0.88, rain x0.6), not a road graph; riders return to the hub between trips.	`src/last_mile_dispatch/ config.py:13, :36; geo.py:29-46; simulate.py:247-249; README 'Known limitations'`	F-32, F-33, F-34, F-35, F-36, F-37, F-39, F-44, F-45, F-46, F-47	slides 06, 07, 08
A-04	Matching prices each rider-bag pair in AED: km cost of the vehicle, 1.0 per minute late, minus 1.4 per minute already waited, minus a 9.0 credit per order. The matcher is myopic (one 2-minute wave at a time).	`src/last_mile_dispatch/ config.py:74-76; assignment.py:46-58; README 'Known limitations'`	F-14, F-33, F-34, F-39, F-44, F-45, F-46, F-47, F-48	slides 03, 06, 07, 08
A-05	Bagging limits: drops within 1 km and created within 12 minutes of the seed order, at most 3 stops, and the first customer may wait at most 9 extra minutes.	`src/last_mile_dispatch/ config.py:37, :65-67; bagging.py:51-87`	F-15, F-16, F-17, F-39, F-41	slides 04, 05, 07
A-06	Geofence rule: a 'customer unavailable' mark is flagged when fewer than 2 of the rider's last 6 GPS pings are within 150 m of the customer; flagged riders are sent back to the door (treatment) or the flag is only logged (baseline).	`src/last_mile_dispatch/ config.py:43, :79-80; geofence.py:31-39; simulate.py:218-230`	F-21, F-25, F-49, F-50	slides 05, 08
A-07	Fake marks: about 8% of riders (36 of 473 in the data) mark 'customer unavailable' without reaching the door on 22% of their stops; fakes are known by construction and riders do not change habit when flagged.	`src/last_mile_dispatch/ config.py:40-41; scripts/generate_data.py:94; README 'Known limitations'`	F-10, F-21, F-49, F-50	slides 02, 05, 08
A-08	Service promise 30 minutes from order creation; a customer cancels if nothing has left the hub after 45 minutes; the dispatcher wakes every 2 minutes; 2 minutes handover per stop.	`src/last_mile_dispatch/ config.py:23, :27-29`	F-07, F-08, F-24, F-32, F-33, F-34, F-42, F-46, F-47, F-48	slides 02, 03, 05, 06, 07, 08
A-09	Experiment design: hubs sorted by pre-period orders per day, paired, one coin flip per pair (seed 2026); one number per hub over four weeks; Welch t and hub-bootstrap 95% intervals; power from the pre-period between-hub spread.	`src/last_mile_dispatch/ cli.py:22, :58-98; experiment.py:20-124`	F-27, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-52, F-53, F-54, F-55, F-56, F-57	slides 05, 08, 09
A-10	Calendar: two pre-period weeks from 2026-01-19, then four experiment weeks 2026-02-02 to 2026-03-01; weekends are Saturday-Sunday; Ramadan from 2026-02-18; one rain day 2026-02-07 (35% more orders, 40% slower traffic).	`src/last_mile_dispatch/ config.py:16-18; scripts/generate_data.py:35-36; README 'The data'`	F-06, F-09, F-13, F-31, F-32, F-35, F-36, F-37, F-38, F-40	slides 02, 03, 05, 06, 07
A-11	The rider board (slide 07) re-runs the repo's own simulator for hub 18 on 2026-02-14 under the treatment and records the 21:46 wave; drop ETAs replay the simulator's clock with geo.travel_min for riders who are honest and customers who are home.	`docs/demo/source/tools/ export_rider_board.py; simulate.py:189-253`	F-39, F-40, F-41, F-42	slides 07
A-12	Video rounding: percentages to 1 decimal, AED to 2 decimals, utilisation and CI bounds to 3 decimals, counts with thousands separators; '~3 min' and '180 s' are rounded run times.	`this audit (display convention)`	F-16, F-17, F-23, F-32, F-34, F-35, F-36, F-37, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-53, F-54, F-55, F-59	slides 04, 05, 06, 08, 09, 10
A-13	Slide 01 (industries, everyday problems) is general delivery-operations knowledge used as context. It is not a result of this repo and carries no numbers.	`author's framing; README 'The problem' supports problems 1-4`	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every few minutes, a hub decides who carries what.

WHO LIVES WITH THIS · Quick-commerce grocery · Food & restaurant delivery · Parcel & pharmacy couriers

01 One order per trip, dispatched by hand — Two orders to one building leave with two riders

02 Peaks outrun the riders — Weekends, holidays and bad weather turn queues into cancellations

03 "Customer unavailable" marked from the road — Each one costs a refund and a customer

04 Changes judged by a before-and-after chart — The calendar moves the numbers more than the change

THIS PROJECT tackles all four, end to end · bag → match → check → measure

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem', 'Insights and learning'` · README text	-	✓

Assumptions: A-13

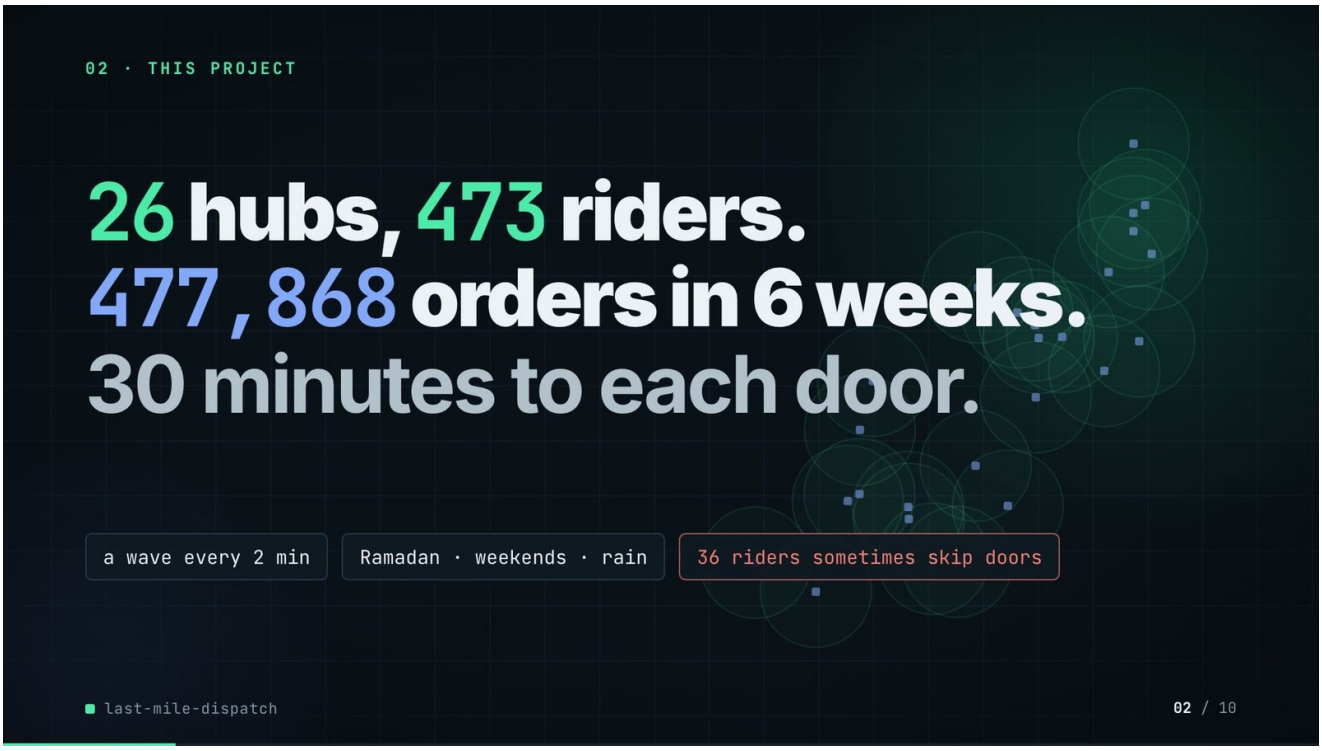
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html` (SERIES block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

02 · THIS PROJECT

26 hubs, 473 riders. 477,868 orders in 6 weeks. 30 minutes to each door.

a wave every 2 min · Ramadan · weekends · rain · 36 riders sometimes skip doors

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	26 hubs	`outputs/summary.json; data/hubs.csv` · hubs; row count	-	✓
F-04	473 riders	`data/riders.csv` · row count	-	✓
F-05	477,868 orders	`data/orders.csv.gz` · row count	-	✓
F-06	6 weeks / 42 days	`data/days.csv` · row count = 42	42 / 7 = 6 weeks	✓
F-07	30 minutes to each door / 30-minute promise	`src/last_mile_dispatch/config.py:27` · SLA_MIN = 30	-	✓
F-08	a wave every 2 min / EVERY 2 MINUTES / dispatch wave 2 min	`src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2	-	✓
F-09	Ramadan · weekends · rain	`data/days.csv` · ramadan, weekend, rain columns	any ramadan, any weekend, exactly one rain day	✓
F-10	36 riders sometimes skip doors	`data/riders.csv` · remote_marker == True	count	✓
F-11	hub dots + circles	`data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0)` · lat/lon -> km from the centroid (demo assets/run_series.json)	-	✓

Assumptions: A-01, A-07, A-08, A-10

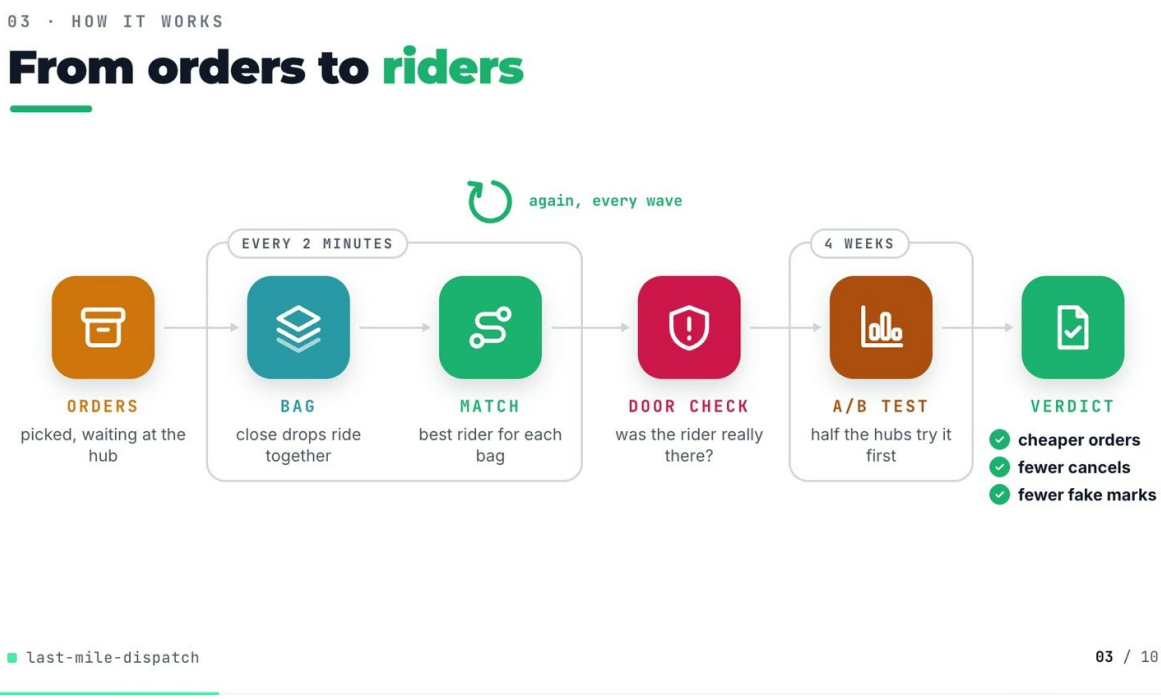
Illustrative, not data:

- The background is the 26 real hub positions with a 4 km service circle each (decoration, low opacity).

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; counts come from data/*.csv.

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From orders to riders

ORDERS picked, waiting at the hub → [EVERY 2 MINUTES: BAG close drops ride together → MATCH best rider for each bag] → DOOR CHECK was the rider really there? → [4 WEEKS: A/B TEST half the hubs try it first] → VERDICT ✓ cheaper orders ✓ fewer cancels ✓ fewer fake marks

again, every wave (loop over MATCH)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-08	a wave every 2 min / EVERY 2 MINUTES / dispatch wave 2 min	`src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2	-	✓
F-12	ORDERS → BAG → MATCH → DOOR CHECK → A/B TEST → VERDICT	`src/last_mile_dispatch/` · functions	-	✓
F-13	4 WEEKS	`outputs/summary.json` · experiment_days = 28	28 / 7 = 4	✓
F-14	cheaper orders / fewer cancels / fewer fake marks	`outputs/ab_readout.csv` · welch_hi for cost_per_delivered, cancel_rate, fake_accepted_per_1k	welch_hi < 0 for all three	✓

Assumptions: A-05, A-06, A-08, A-09

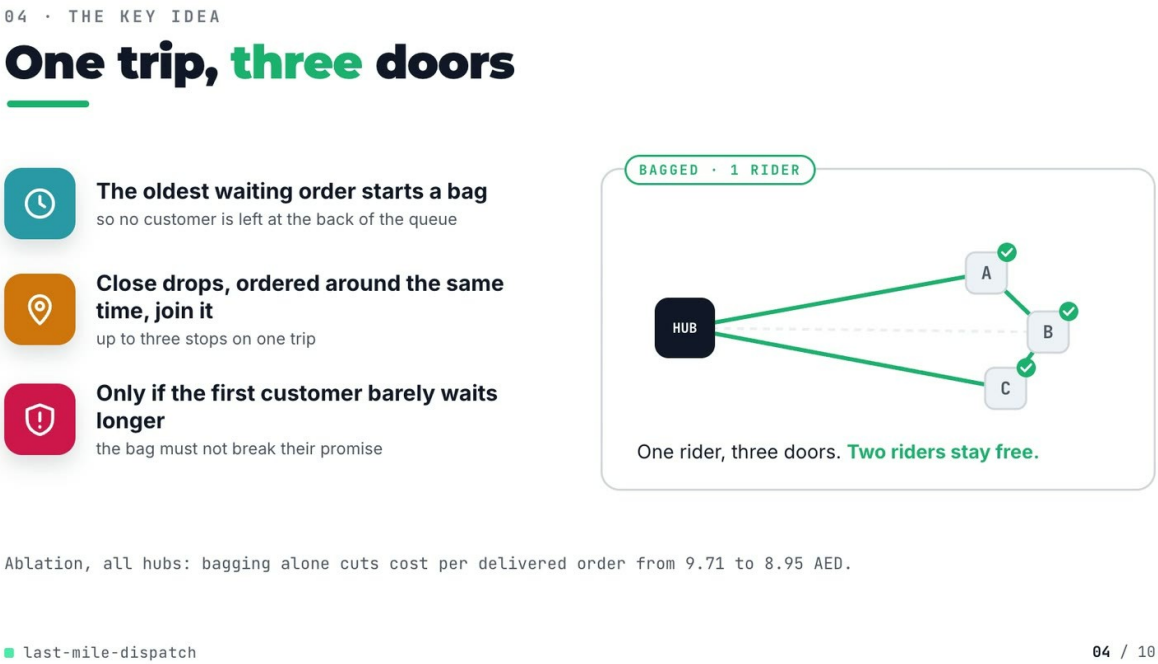
Illustrative, not data:

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from ab_readout.csv (F-14).

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

One trip, three doors

The oldest waiting order starts a bag — so no customer is left at the back of the queue

Close drops, ordered around the same time, join it — up to three stops on one trip

Only if the first customer barely waits longer — the bag must not break their promise

ONE ORDER PER TRIP · 3 RIDERS → BAGGED · 1 RIDER (HUB, A, B, C). One rider, three doors. Two riders stay free.

Ablation, all hubs: bagging alone cuts cost per delivered order from 9.71 to 8.95 AED.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-15	3 steps / ≤ 3 stops · 1 km	`src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67` · BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0	-	✓
F-16	9.71	`outputs/ablation.csv` · policy=baseline, cost_per_delivered	-	✓
F-17	8.95 AED	`outputs/ablation.csv` · policy=bagging, cost_per_delivered	-	✓

Assumptions: A-02, A-05, A-12

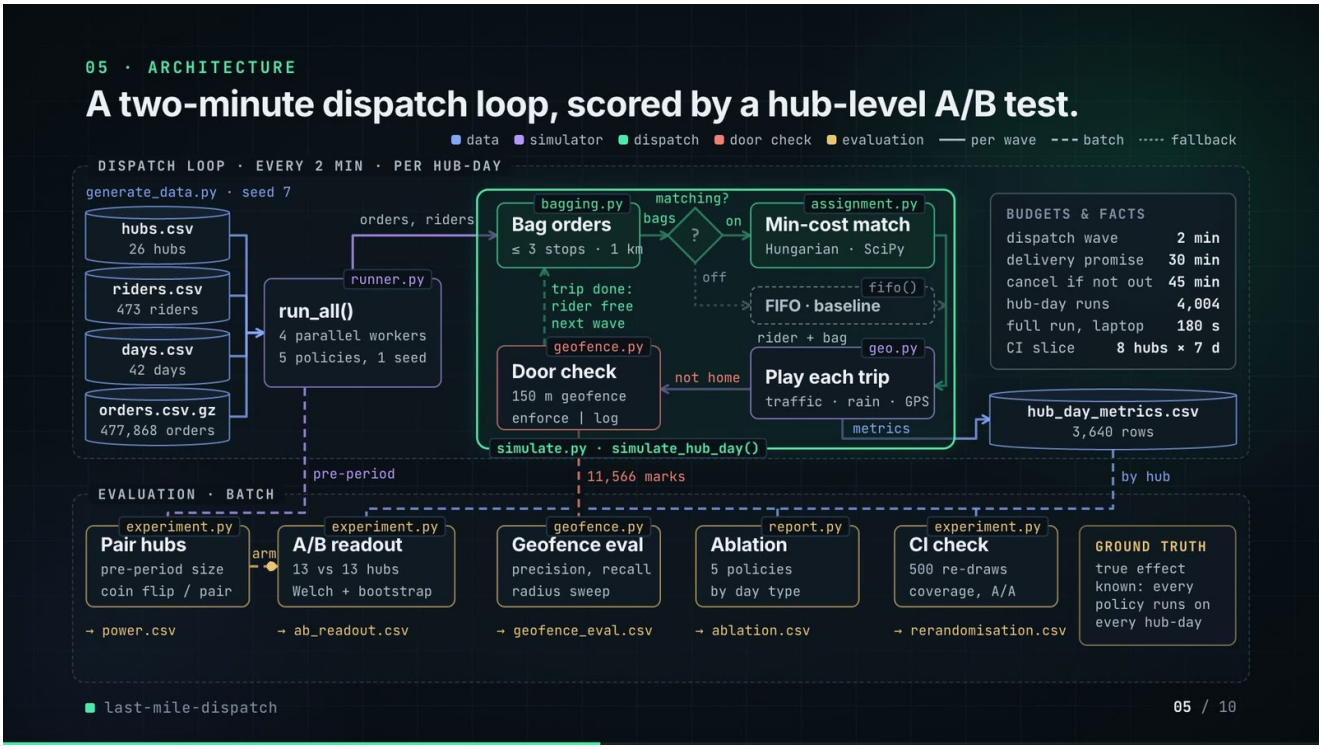
Illustrative, not data:

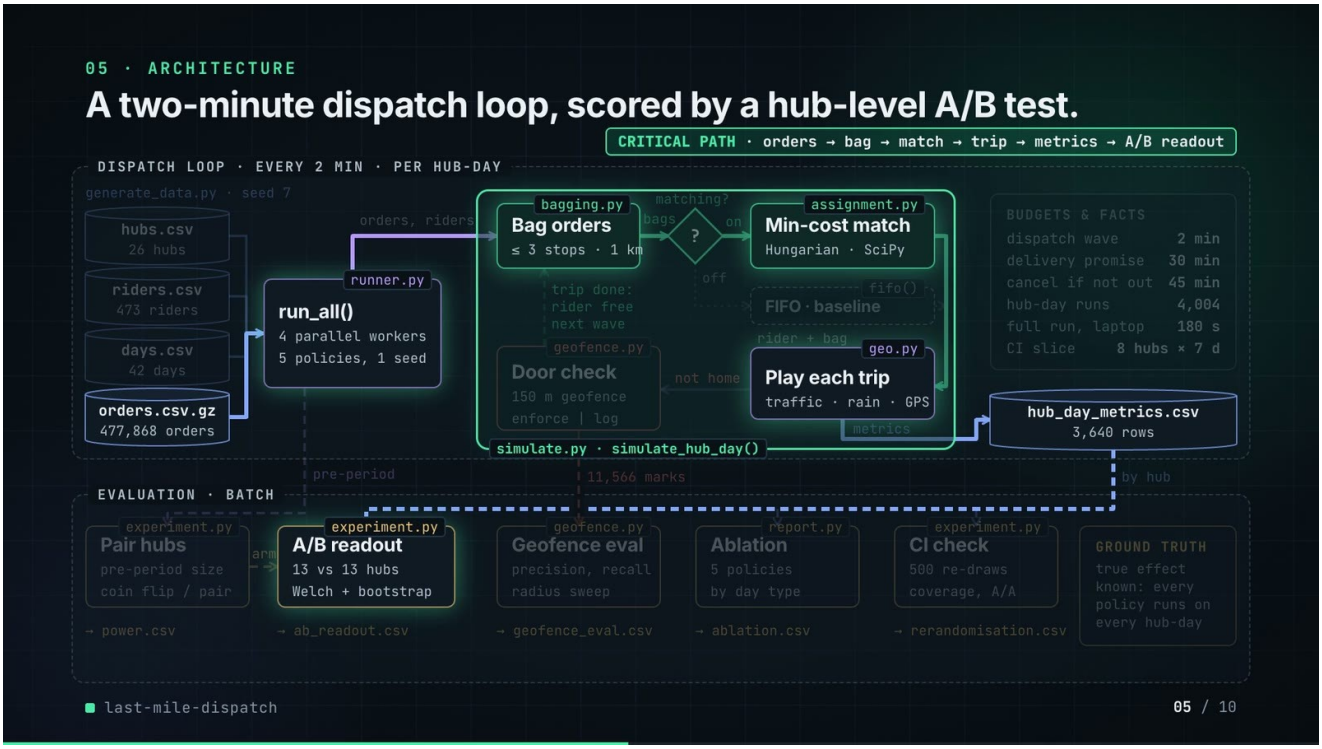
- The HUB/A/B/C demo is schematic: it shows the rule in bagging.make_bags, not a real bag.

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-16 and F-17.

Slide 05: Architecture

26.0-41.0 s · composition `compositions/s03-architecture.html`





On-screen text (verbatim)

05 · ARCHITECTURE

A two-minute dispatch loop, scored by a hub-level A/B test.

Legend: data · simulator · dispatch · door check · evaluation · per wave (solid) · batch (dashed) · fallback (dotted)

DISPATCH LOOP · EVERY 2 MIN · PER HUB-DAY — generate_data.py · seed 7: hubs.csv 26 hubs, riders.csv 473 riders, days.csv 42 days, orders.csv.gz 477,868 orders

runner.py run_all(): 4 parallel workers, 5 policies, 1 seed → orders, riders →

simulate.py · simulate_hub_day(): bagging.py Bag orders (≤ 3 stops · 1 km) → bags → matching? → on → assignment.py Min-cost match (Hungarian · SciPy); off → fifo() FIFO · baseline → rider + bag → geo.py Play each trip (traffic · rain · GPS) → not home → geofence.py Door check (150 m geofence · enforce | log) → trip done: rider free, next wave

metrics → hub_day_metrics.csv 3,640 rows · BUDGETS & FACTS: dispatch wave 2 min, delivery promise 30 min, cancel if not out 45 min, hub-day runs 4,004, full run, laptop 180 s, CI slice 8 hubs × 7 d

EVALUATION · BATCH: experiment.py Pair hubs (pre-period size, coin flip / pair) → arm → experiment.py A/B readout (13 vs 13 hubs, Welch + bootstrap) → ab_readout.csv; geofence.py Geofence eval (precision, recall, radius sweep) ← 11,566 marks → geofence_eval.csv; report.py Ablation (5 policies, by day type) → ablation.csv; experiment.py CI check (500 re-draws, coverage, A/A) → rerandomisation.csv; → power.csv; GROUND TRUTH: true effect known: every policy runs on every hub-day

CRITICAL PATH · orders → bag → match → trip → metrics → A/B readout

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	26 hubs	`outputs/summary.json; data/hubs.csv` · hubs; row count	-	✓
F-04	473 riders	`data/riders.csv` · row count	-	✓
F-05	477,868 orders	`data/orders.csv.gz` · row count	-	✓
F-06	6 weeks / 42 days	`data/days.csv` · row count = 42	42 / 7 = 6 weeks	✓
F-08	a wave every 2 min / EVERY 2 MINUTES / dispatch wave 2 min	`src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2	-	✓
F-15	3 steps / ≤ 3 stops · 1 km	`src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67` · BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0	-	✓
F-18	generate_data.py, runner.py run_all(), simulate.py simulate_hub_day(), bagging.py, assignment.py (min_cost, fifo), geo.py, geofence.py, experiment.py, report.py	`src/last_mile_dispatch/, scripts/` · defs	-	✓
F-19	power.csv, ab_readout.csv, geofence_eval.csv, ablation.csv, rerandomisation.csv, hub_day_metrics.csv	`outputs/` · files	-	✓
F-20	3,640 rows	`outputs/hub_day_metrics.csv` · row count	26 x 28 x 5	✓
F-21	11,566 marks	`outputs/geofence_eval.csv` · slice=all, events	-	✓
F-22	hub-day runs 4,004	`outputs/run_log.txt line 2` · 'simulated 4,004 hub-day-policy runs'	26 x 14 + 26 x 28 x 5 = 4,004	✓
F-23	full run, laptop 180 s	`outputs/summary.json` · runtime_s = 179.8	rounded to whole seconds	✓
F-24	cancel if not out 45 min	`src/last_mile_dispatch/config.py:28` · CANCEL_AFTER_MIN = 45	-	✓
F-25	150 m geofence	`src/last_mile_dispatch/config.py:79` · GEOFENCE_RADIUS_M = 150.0	-	✓
F-26	500 re-draws	`outputs/rerandomisation.csv` · draws	-	✓
F-27	13 vs 13 hubs	`outputs/summary.json` · arms.treatment, arms.control	len of each list	✓
F-28	4 parallel workers · 5 policies, 1 seed	`src/last_mile_dispatch/cli.py:29; runner.py:36-41; config.py:62` · --workers default 4; same seed for every policy; ALL_POLICIES	-	✓
F-29	CI slice 8 hubs × 7 d	`src/last_mile_dispatch/cli.py:38, :40; .github/workflows/ci.yml` · hubs.head(8), experiment days head(7)	-	✓
F-30	generate_data.py · seed 7	`scripts/generate_data.py:168` · --seed default 7	-	✓

Assumptions: A-01, A-03, A-04, A-05, A-06, A-08, A-09

Illustrative, not data:

- Packets moving along the wires are decoration in execution order (the dispatch loop turns twice).
- 'per wave' (solid) vs 'batch' (dashed) marks what runs inside each 2-minute wave and what runs after all hub-days.
- The '?' diamond is the decision point Policy.matching (config.py); the dotted path is the FIFO fallback used by the baseline.
- The critical-path highlight is the chain that produces the headline A/B numbers.

To change this slide: Node boxes (#n-*), cylinders (CYL) and the W wires object in compositions/s03-architecture.html.

Slide 06: The system running

40.6-53.2 s · composition `compositions/s04-routes.html` (SERIES block injected by `tools/inject_sheet.py` from `tools/export_series.py`)





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

Four weeks, 26 hubs, same orders, two dispatchers.

Share of orders cancelled, per day, all hubs · FIFO, one order per trip vs bag + match + geofence · RAMADAN band · RAIN DAY · 42.5% → 14.0%

CANCELLED · 4 WEEKS: FIFO 37,530 · bag+match+geofence 11,063 · −70.5%

WHERE THE SAVING COMES FROM · COST PER DELIVERED ORDER, AED: Ordinary weekday (125,715 orders) 9.71 → 9.36 −3.7% · Weekend or Ramadan (184,449 orders) 9.65 → 8.30 −14.0% · Rain day (16,817 orders) 10.42 → 6.84 −34.4%

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-31	two lines, Feb 2 - Mar 1	outputs/hub_day_metrics.csv · sum(cancelled) / sum(orders) by date and policy	injected SERIES block must equal the recomputation	✓
F-32	RAIN DAY · 42.5% → 14.0%	outputs/stress_split.csv · day_type=rain day, cancel_rate	x 100	✓
F-33	37,530 → 11,063	outputs/hub_day_metrics.csv · sum(cancelled) by policy	-	✓
F-34	−70.5%	outputs/hub_day_metrics.csv · F-33	(11,063 / 37,530 - 1) x 100 = -70.52%	✓
F-35	125,715 orders · 9.71 → 9.36 · −3.7%	outputs/stress_split.csv · day_type=ordinary weekday	cost_treat / cost_base - 1	✓
F-36	184,449 orders · 9.65 → 8.30 · −14.0%	outputs/stress_split.csv · day_type=weekend or Ramadan	cost_treat / cost_base - 1	✓
F-37	16,817 orders · 10.42 → 6.84 · −34.4%	outputs/stress_split.csv · day_type=rain day	cost_treat / cost_base - 1	✓
F-38	RAMADAN band from Feb 18	data/days.csv · first date with ramadan = True	-	✓

Assumptions: A-02, A-03, A-08, A-10

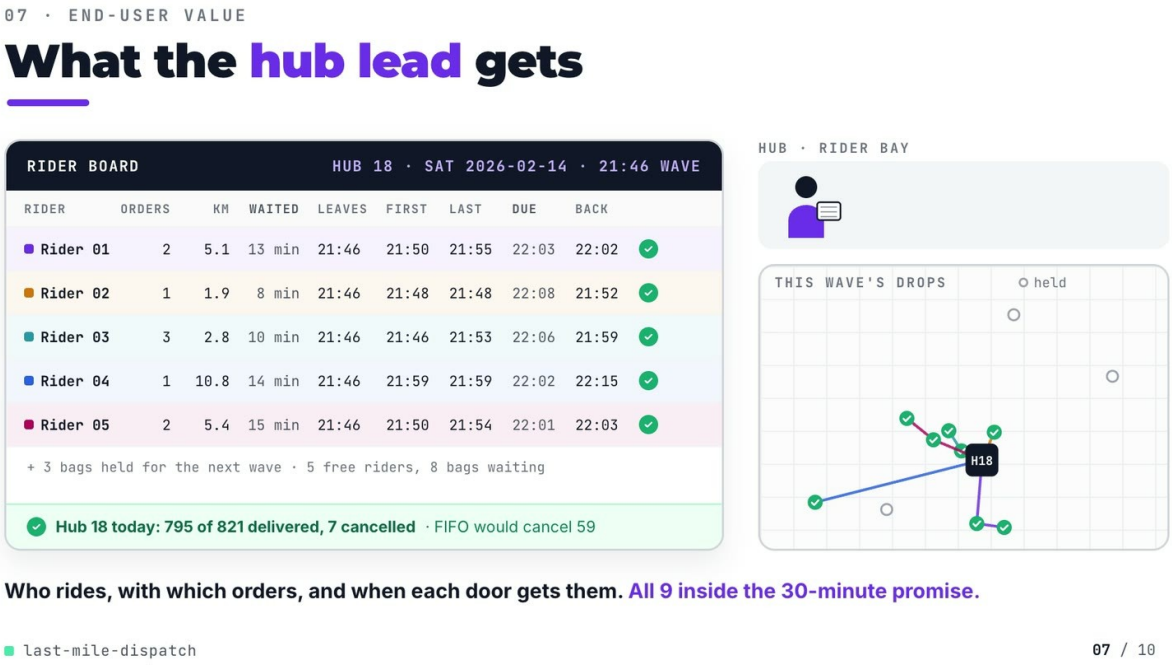
Illustrative, not data:

- Chart and bars are drawn from the injected numbers (no hand-typed data points); bars are to scale from zero.

To change this slide: Re-run tools/export_series.py and inject it (--name SERIES); layout in compositions/s04-routes.html.

Slide 07: End-user value (white)

52.9-65.9 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py` from `tools/export_rider_board.py`)



On-screen text (verbatim)

07 · END-USER VALUE
What the hub lead gets
RIDER BOARD · HUB 18 · SAT 2026-02-14 · 21:46 WAVE — columns RIDER, ORDERS, KM, WAITED, LEAVES, FIRST, LAST, DUE, BACK
Rider 01 2 5.1 13 min 21:46 21:50 21:55 22:03 22:02 ✓
Rider 02 1 1.9 8 min 21:46 21:48 21:48 22:08 21:52 ✓
Rider 03 3 2.8 10 min 21:46 21:46 21:53 22:06 21:59 ✓
Rider 04 1 10.8 14 min 21:46 21:59 21:59 22:02 22:15 ✓
Rider 05 2 5.4 15 min 21:46 21:50 21:54 22:01 22:03 ✓
+ 3 bags held for the next wave · 5 free riders, 8 bags waiting
✓ Hub 18 today: 795 of 821 delivered, 7 cancelled · FIFO would cancel 59
HUB · RIDER BAY · THIS WAVE'S DROPS · held
Who rides, with which orders, and when each door gets them. All 9 inside the 30-minute promise.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	26 hubs	`outputs/summary.json; data/hubs.csv` · hubs; row count	-	✓
F-07	30 minutes to each door / 30-minute promise	`src/last_mile_dispatch/config.py:27` · SLA_MIN = 30	-	✓
F-39	Rider 01-05 rows	`demo assets/rider_board.json (re-run of the repo's simulator), injected into compositions/value.html` · rows	km = sum of road legs; waited = wave time - oldest order's creation; due = earliest creation + 30 min	✓
F-40	795 of 821 delivered, 7 cancelled · FIFO would cancel 59	`outputs/hub_day_metrics.csv` · hub_id=18, date=2026-02-14, policy in {bagging+matching+geofence, baseline}	-	✓
F-41	+ 3 bags held for the next wave · 5 free riders, 8 bags waiting	`demo assets/rider_board.json` · bags_held, free_riders, bags_waiting_before	held = bags waiting - riders matched	✓
F-42	All 9 inside the 30-minute promise.	`demo assets/rider_board.json` · orders_dispatched; stops[].on_time	every stop ETA - creation <= 30 min	✓
F-43	HUB 18	`outputs/summary.json` · arms.treatment	-	✓

Assumptions: A-03, A-04, A-05, A-08, A-11

Illustrative, not data:

- Rider 01-05 are placeholders (no real people); all five are on bikes.
- The hub-lead figure, rider bay, scooter drawings and driving speed are illustrative; drop positions and stop order are real, scaled to fit the panel (uniform scale).
- Grey circles are the drops of the 3 bags held for the next wave.

To change this slide: Re-run tools/export_rider_board.py (hub, date, wave arguments), then tools/inject_sheet.py --name SHEET; layout in compositions/value.html.

Slide 08: Results

65.6-74.4 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · HUB-LEVEL A/B, 4 WEEKS

13 treated hubs vs 13 control hubs: cheaper and faster.

COST −10.6% fleet cost per delivered order · AED 9.79 → 8.75

PROMISE +28.3 points more orders inside 30 min · 58.0% → 86.2%

CANCELLATIONS 3.7% cancellation rate, treated hubs · control 11.6%

DOOR CHECK −72.4% fake "not home" marks accepted · 11.7 → 3.2 per 1,000

326,981 orders · Welch 95% CIs exclude zero on all four · true effect known from the simulator

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-27	13 vs 13 hubs	`outputs/summary.json` · arms.treatment, arms.control	len of each list	✓
F-44	−10.6%	`outputs/ab_readout.csv` · metric=cost_per_delivered, rel_diff	x 100	✓
F-45	AED 9.79 → 8.75	`outputs/ab_readout.csv` · metric=cost_per_delivered, control, treatment	-	✓
F-46	+28.3	`outputs/ab_readout.csv` · metric=on_time_share_of_orders, diff	x 100	✓
F-47	58.0% → 86.2%	`outputs/ab_readout.csv` · metric=on_time_share_of_orders, control, treatment	x 100	✓
F-48	3.7% · control 11.6%	`outputs/ab_readout.csv` · metric=cancel_rate, treatment, control	x 100	✓
F-49	−72.4%	`outputs/ab_readout.csv` · metric=fake_accepted_per_1k, rel_diff	x 100	✓
F-50	11.7 → 3.2 per 1,000	`outputs/ab_readout.csv` · metric=fake_accepted_per_1k, control, treatment	-	✓
F-51	326,981 orders	`outputs/summary.json` · orders_in_experiment	-	✓
F-52	Welch 95% CIs exclude zero on all four	`outputs/ab_readout.csv` · welch_lo, welch_hi	lo > 0 or hi < 0 for cost, on-time, cancel, fake	✓

Assumptions: A-02, A-03, A-04, A-06, A-09, A-12

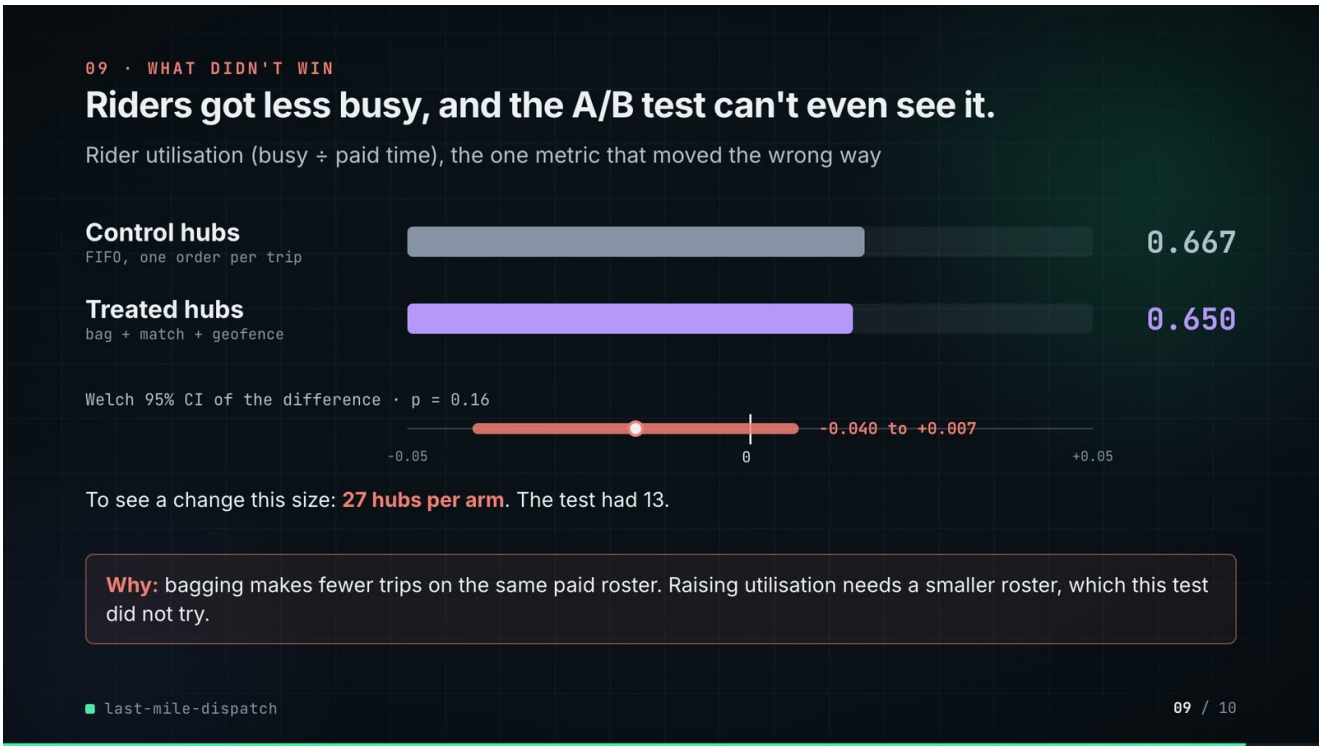
Illustrative, not data:

- Mini bars under each number are drawn to the same ratio (treatment / control).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

74.0-80.6 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

Riders got less busy, and the A/B test can't even see it.

Rider utilisation (busy ÷ paid time), the one metric that moved the wrong way

Control hubs (FIFO, one order per trip) 0.667 · Treated hubs (bag + match + geofence) 0.650

Welch 95% CI of the difference · p = 0.16 · −0.040 to +0.007 (axis −0.05 … 0 … +0.05)

To see a change this size: 27 hubs per arm. The test had 13.

Why: bagging makes fewer trips on the same paid roster. Raising utilisation needs a smaller roster, which this test did not try.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-27	13 vs 13 hubs	`outputs/summary.json` · arms.treatment, arms.control	len of each list	✓
F-53	0.667 / 0.650	`outputs/ab_readout.csv` · metric=utilization, control, treatment	-	✓
F-54	−0.040 to +0.007	`outputs/ab_readout.csv` · metric=utilization, welch_lo, welch_hi	-	✓
F-55	p = 0.16	`outputs/ab_readout.csv` · metric=utilization, welch_p	-	✓
F-56	27 hubs per arm	`outputs/power.csv` · metric=utilization, hubs_per_arm_needed	-	✓
F-57	axis −0.05..+0.05 = 1,000 px; bar 95 → 571 px, estimate at 333 px	`compositions/s06-twist.html`; `outputs/ab_readout.csv` · welch_lo, welch_hi, diff	px = (v + 0.05) x 10,000	✓

Assumptions: A-02, A-09, A-12

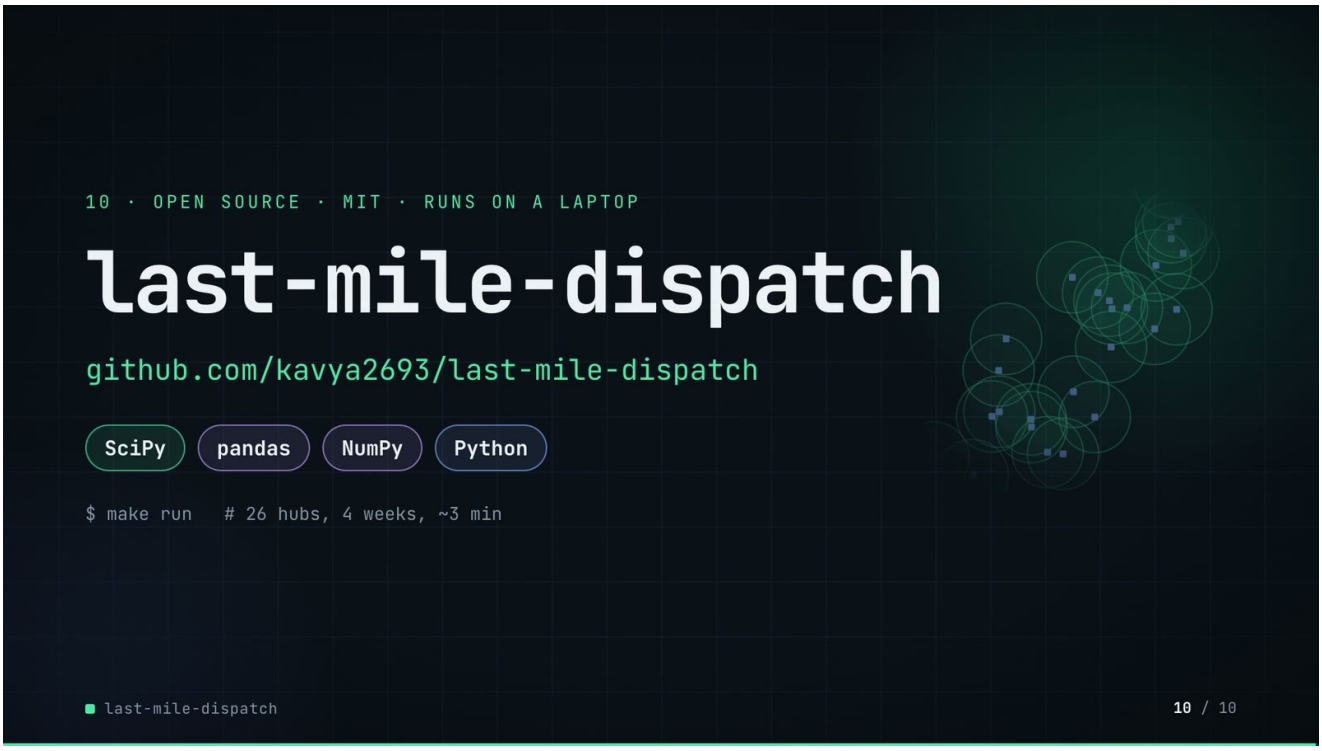
Illustrative, not data:

- Utilisation bars: 1,000 px track = 1.0 (from zero).

To change this slide: Rows #twist-r1..r2 and the CI strip (#twist-ci) in compositions/s06-twist.html.

Slide 10: End card

80.2-84.6 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP

last-mile-dispatch

github.com/kavya2693/last-mile-dispatch

SciPy · pandas · NumPy · Python

$ make run # 26 hubs, 4 weeks, ~3 min

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-11	hub dots + circles	`data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0)` · lat/lon -> km from the centroid (demo assets/run_series.json)	-	✓
F-58	MIT	`LICENSE line 1` · -	-	✓
F-59	~3 min	`outputs/summary.json` · runtime_s = 179.8	179.8 / 60 = 3.0 min	✓
F-60	SciPy · pandas · NumPy · Python	`pyproject.toml dependencies` · numpy, pandas, scipy	-	✓
F-61	$ make run # 26 hubs, 4 weeks, ~3 min	`Makefile; outputs/summary.json` · run target; hubs; experiment_days	-	✓

Assumptions: A-12

Illustrative, not data:

- Faded background: the 26 real hub positions with 4 km circles.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (26 hubs, 473 riders, 42 days, 477,868 orders)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`scripts/generate_data.py`; `src/last_mile_dispatch/config.py`	make data; make run
A-02: The rider roster is fixed and sized for an ordinary weekday evening; every policy pays for the same riders	F-04, F-33, F-34, F-35, F-36, F-37, F-44, F-45, F-53, F-54, F-55, F-56	02, 05, 06, 08, 09	`scripts/generate_data.py:84-85`	make data; make run
A-03: Travel is straight-line distance x 1	F-32, F-33, F-34, F-35, F-36, F-37, F-39, F-44, F-45, F-46, F-47	06, 07, 08	`src/last_mile_dispatch/config.py:13, :36`; `src/last_mile_dispatch/geo.py:29-41`	make run; export_rider_board.py + inject_sheet.py
A-04: Matching prices each rider-bag pair in AED: km cost of the vehicle, 1	F-14, F-33, F-34, F-39, F-44, F-45, F-46, F-47, F-48	03, 06, 07, 08	`src/last_mile_dispatch/config.py:74-76`	make run; export_rider_board.py + inject_sheet.py
A-05: Bagging limits: drops within 1 km and created within 12 minutes of the seed order, at most 3 stops, and the first customer may wait at most 9 extra minutes	F-15, F-16, F-17, F-39, F-41	04, 05, 07	`src/last_mile_dispatch/config.py:37, :65-67`	make run; export_rider_board.py + inject_sheet.py
A-06: Geofence rule: a 'customer unavailable' mark is flagged when fewer than 2 of the rider's last 6 GPS pings are within 150 m of the customer; flagged riders are sent back to the door (treatment) or the flag is only logged (baseline)	F-21, F-25, F-49, F-50	05, 08	`src/last_mile_dispatch/config.py:79-80`	make run
A-07: Fake marks: about 8% of riders (36 of 473 in the data) mark 'customer unavailable' without reaching the door on 22% of their stops; fakes are known by construction and riders do not change habit when flagged	F-10, F-21, F-49, F-50	02, 05, 08	`src/last_mile_dispatch/config.py:40-41`	make data; make run
A-08: Service promise 30 minutes from order creation; a customer cancels if nothing has left the hub after 45 minutes; the dispatcher wakes every 2 minutes; 2 minutes handover per stop	F-07, F-08, F-24, F-32, F-33, F-34, F-42, F-46, F-47, F-48	02, 03, 05, 06, 07, 08	`src/last_mile_dispatch/config.py:23-29`	make run; export_rider_board.py + inject_sheet.py
A-09: Experiment design: hubs sorted by pre-period orders per day, paired, one coin flip per pair (seed 2026); one number per hub over four weeks; Welch t and hub-bootstrap 95% intervals; power from the pre-period between-hub spread	F-27, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-52, F-53, F-54, F-55, F-56, F-57	05, 08, 09	`src/last_mile_dispatch/cli.py:22`; `src/last_mile_dispatch/experiment.py`	make run
A-10: Calendar: two pre-period weeks from 2026-01-19, then four experiment weeks 2026-02-02 to 2026-03-01; weekends are Saturday-Sunday; Ramadan from 2026-02-18; one rain day 2026-02-07 (35% more orders, 40% slower traffic)	F-06, F-09, F-13, F-31, F-32, F-35, F-36, F-37, F-38, F-40	02, 03, 05, 06, 07	`src/last_mile_dispatch/config.py:16-18`; `scripts/generate_data.py:35-36`	make data; make run
A-11: The rider board (slide 07) re-runs the repo's own simulator for hub 18 on 2026-02-14 under the treatment and records the 21:46 wave; drop ETAs replay the simulator's clock with geo	F-39, F-40, F-41, F-42	07	`docs/demo/source/tools/export_rider_board.py (hub, date, wave arguments)`	export_rider_board.py + inject_sheet.py; hyperframes render
A-12: Video rounding: percentages to 1 decimal, AED to 2 decimals, utilisation and CI bounds to 3 decimals, counts with thousands separators; '~3 min' and '180 s' are rounded run times	F-16, F-17, F-23, F-32, F-34, F-35, F-36, F-37, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-53, F-54, F-55, F-59	04, 05, 06, 08, 09, 10	`the composition file named on each slide`	hyperframes render
A-13: Slide 01 (industries, everyday problems) is general delivery-operations knowledge used as context	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

• Everything is simulated (A-01). The A/B readout is what an experimenter would see; the 'true effect' is known only because every policy ran on every hub-day with the same random draws.

• The cost saving is mostly cancellations turned into deliveries on a fixed, paid roster (A-02); slide 06 shows it is under 4% on an ordinary weekday and 34% on the rain day.

• Slide 07's ETAs are reconstructed with the repo's travel function for one wave; they equal the simulator's delivery times because the five riders are honest and all nine customers are home (asserted in export_rider_board.py).

• Slide 08 uses hub-level means (13 vs 13); the ablation numbers on slide 04 pool every hub and every day, so they differ slightly from the A/B means.

• '36 riders sometimes skip doors' is a generator setting (8% of riders, 22% of their stops), not an observed behaviour (A-07).

• Slide 01 statements are general industry knowledge (A-13).

• Rider names on slide 07 are placeholders.

Appendix: environment

Item	Value
Commit	`c7de8a6ad78ef3961f773e55ae9af197085c3f9c`
Python	3.11.7
numpy	2.4.6
pandas	3.0.6
scipy	1.17.1
Profile / seeds	data=7, hub_assignment=2026, rerandomisation=2027, simulator=hub_id * 10000 + day index (same for every policy)
Run time	summary.json runtime_s = 179.8 s (about 3 min) for make run with 4 workers on the author's laptop